

ORIGINAL ARTICLE

Get With The Guidelines–Heart Failure Hospital Participation and Its Association With Guideline-Directed Medical Therapy and Outcomes

Aradhana Verma¹ MD, MTM; Gregg C. Fonarow¹ MD; Paul Heidenreich¹ MD; Larry A. Allen¹ MD, MHS; Andrew P. Ambrosy¹ MD; Shun Kohsaka¹ MD, PhD; Anubodh S. Varshney¹ MD; Nicholas Brownwell¹ MD, PhD; Jun Fan, MS; Alexander T. Sandhu¹ MD, MS

BACKGROUND: Despite strong evidence, adoption of guideline-directed medical therapy (GDMT) for heart failure with reduced ejection fraction remains suboptimal. The Get With The Guidelines–Heart Failure (GWTG-HF) program was designed to close gaps in care. We evaluated whether hospital participation in GWTG-HF was associated with greater GDMT intensity and improved outcomes.

METHODS: We conducted a retrospective analysis (2013–2021) of Medicare beneficiaries with part A and part D hospitalized with heart failure with reduced ejection fraction. Using a multiple baseline time series design, we compared changes in GDMT and outcomes before and after GWTG-HF enrollment with hospitals that never participated. Coprimary outcomes were a 90-day postdischarge GDMT score under a parallel and nonparallel slopes model. Secondary outcomes included class-specific medication fills, achievement of $\geq 50\%$ target doses, and 30-day, 90-day, and 1-year all-cause and HF readmission and mortality. Mortality and first HF readmission were also evaluated using Cox proportional hazards models. We adjusted for baseline hospital performance, patient characteristics, and temporal trends.

RESULTS: Among 1 274 863 Medicare beneficiaries hospitalized for heart failure with reduced ejection fraction, 53.5% were treated at hospitals that never participated in GWTG-HF and 9.6% at GWTG-HF hospitals. Unadjusted median GDMT scores increased from 3.0 in both groups to 4.0 in nonparticipating hospitals and 4.5 in GWTG-HF hospitals at 90 days ($P < 0.001$). Hospital enrollment was associated with a higher 90-day GDMT score ($+0.15$ points [95% CI, 0.12–0.20]; $P < 0.001$) and greater use of β -blockers, renin-angiotensin system inhibitors, and mineralocorticoid receptor antagonists but not angiotensin receptor-neprilysin inhibitors. GWTG-HF participation was associated with lower all-cause mortality at 30 days (odds ratio, 0.95 [95% CI, 0.92–0.98]) and 1 year (0.97 [95% CI, 0.95–0.1.00]; both $P < 0.05$). Mortality differences were attenuated and no longer significant in the nonparallel slopes model.

CONCLUSIONS: Hospital participation in GWTG-HF was associated with modest but significant improvements in postdischarge GDMT intensity, supporting the value of quality improvement initiatives to address persistent treatment gaps in heart failure with reduced ejection fraction.

Key Words: heart failure ■ hospitals ■ interrupted time series analysis ■ Medicare ■ quality improvement

The Get With The Guidelines–Heart Failure (GWTG-HF) registry is a national, hospital-based quality improvement initiative launched in 2005 to enhance heart failure (HF) care by promoting adherence to accepted

treatment guidelines.^{1,2} The program includes workshops, webinars, hospital toolkits, performance benchmarks, and achievement awards to drive high-quality HF care.³ Although prior analyses demonstrated that GWTG-HF

Correspondence to: Aradhana Verma, MD, MTM, Division of Cardiovascular Medicine, The Cardiovascular Institute, Stanford University School of Medicine, 300 Pasteur Dr, Stanford, CA 94305. Email arverma@stanford.edu

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WHAT IS NEW?

- This study evaluated the association between hospital enrollment in the Get With The Guidelines–Heart Failure program and changes in heart failure care over time using a multiple baseline time series design.
- Using national Medicare data from 2013 to 2021, hospital participation was associated with modest improvements in postdischarge guideline-directed medical therapy, particularly for β-blockers, renin-angiotensin system inhibitors, and mineralocorticoid receptor antagonists.
- The magnitude of improvement was attenuated in analyses allowing for nonparallel preenrollment trends. Similarly, the observed association with lower mortality was attenuated and no longer significant in the alternative model.

WHAT ARE THE CLINICAL IMPLICATIONS?

- Hospital-based quality improvement programs can improve the delivery of evidence-based therapies for patients hospitalized with heart failure with reduced ejection fraction, although the magnitude of improvement was noted to be modest.
- These findings support continued investment in structured quality initiatives such as Get With The Guidelines–Heart Failure while highlighting the need to strengthen outpatient guideline-directed medical therapy optimization, dose titration, and implementation of newer therapies. Expanding participation among under-resourced hospitals and identifying the program components with the greatest impact may further improve the quality of heart failure care.

Nonstandard Abbreviations and Acronyms

ACE	angiotensin-converting enzyme
ARB	angiotensin II receptor blocker
GDMT	guideline-directed medical therapy
GWTHG-HF	Get With The Guidelines–Heart Failure
HF	heart failure
HFrEF	heart failure with reduced ejection fraction
IQR	interquartile range
MRA	mineralocorticoid receptor antagonist
RASI	renin-angiotensin system inhibitor
SMD	standardized mean difference

hospitals have higher performance on process of care measures compared with nonparticipating hospitals, the association between joining the GWTHG-HF registry and subsequent within-hospital changes in HF quality of care and outcomes has not been previously evaluated.⁴

The primary objective of this study was to evaluate the association between hospital enrollment in GWTHG-HF and quality of care for patients with HF with reduced ejection fraction (HFrEF) based on guideline-directed medical therapy (GDMT) utilization and clinical outcomes. Using national Medicare claims data linked to hospital GWTHG-HF participation data, we compared changes in HF quality of care and outcomes over time with a multiple baseline time series analysis between hospitals that joined the GWTHG-HF registry and those that never participated between 2013 and 2021.

METHODS

Study Design

We used a multiple baseline time series design to assess the association between GWTHG-HF participation and outcomes (Figure 1). This quasi-experimental approach is conceptually similar to a controlled interrupted time series analysis but is well-suited for interventions introduced in a staggered fashion across different sites at different time points as compared with a single intervention implemented simultaneously across sites.^{5,6} In this design, each hospital serves as its own comparator before enrollment, although hospitals not yet enrolled and those that never participated in GWTHG-HF during the study period served as concurrent controls, allowing separation of the intervention from underlying secular trends. The staggered enrollment of hospitals into GWTHG-HF at varying time points strengthens the analysis by providing multiple interruption points and reducing bias from time-varying effects common to all sites. This approach has been previously used to evaluate system-level healthcare interventions introduced sequentially across institutions and provides a robust framework for assessing real-world quality improvement initiatives.^{5,7}

Study Population

We analyzed data from Medicare fee-for-service claims from January 1, 2011 to December 31, 2022. We identified hospitalizations for HFrEF between 2013 and 2021 based on a principal discharge diagnosis of HF in a short-term acute-care hospital. Patients were classified as having HFrEF if they had only systolic HF ICD codes during the index hospitalization.⁸ Among participants with systolic and diastolic HF codes during their hospitalization, patients were classified as HFrEF if the majority of HF codes documented in a 1-year lookback period were systolic HF codes.

We restricted the analysis to individuals with continuous enrollment in Medicare Part A for at least 1 year to ensure capture of baseline comorbidities. We also required enrollment in Medicare part D (outpatient prescription plan) for at least 6 months before hospitalization to capture preadmission GDMT and at least 6 months postdischarge or until death.

We excluded patients with a history of heart transplant, left ventricular assist device implantation, advanced chronic kidney disease stage 4 or 5, or dialysis dependence, given these clinical characteristics impact the benefits of GDMT. We also excluded individuals who were transferred to another acute-care short-stay hospital, died during hospitalization, or were discharged to hospice. We included all eligible hospitalizations in

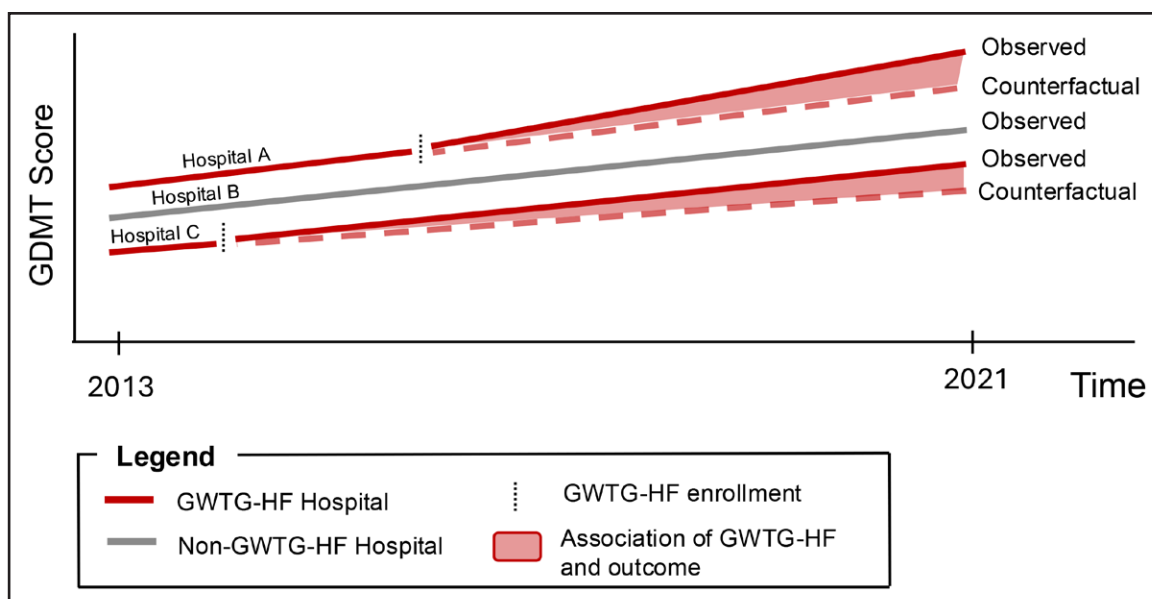


Figure 1. Multiple baseline time series design.

Schematic of the multiple baseline time series design showing staggered Get With The Guidelines–Heart Failure (GWTG-HF) enrollment across hospitals compared with the non-GWTG-HF control arm. Number of hospitals and y axis are representative only and do not reflect actual values. GDMT indicates guideline-directed medical therapy.



the analysis, allowing multiple hospitalizations for an individual Medicare patient. We used cluster-robust standard errors at the patient level to account for within-patient correlation arising from recurrent hospitalizations. The data underlying the findings of this study may be obtained from the corresponding author upon reasonable request.

Exposure

The primary exposure was hospitalization at a facility with active GWTG-HF registry participation. Patients were classified based on the participation status of the admitting hospital at the time of hospitalization. Specifically, patients admitted during periods when their hospital was actively enrolled in the GWTG-HF registry were classified as patients with GWTG-HF, whereas those admitted to hospitals during periods without active registry participation were classified as patients with non-GWTG-HF. Because hospitals enrolled in GWTG-HF at different times throughout the study period, the number of participating hospitals increased over time (Figure S1). Individual hospitals could contribute observations to both exposure groups, contributing patients to the non-GWTG-HF group before enrollment and the GWTG-HF group after enrollment.

We excluded patients treated at hospitals who enrolled in GWTG-HF before the study period, since we lacked data on hospital quality of care before their enrollment in the GWTG-HF registry. We also excluded patients at hospitals who had previously participated in GWTG-HF but disenrolled before the hospitalization.

Outcomes

We defined 2 coprimary outcomes. The primary outcome was a composite GDMT score at 90 days postdischarge reflecting both use and dose intensity of evidence-based therapies for HFrEF, adapted from the continuous 4-Pillar Score.⁹

β -Blockers, ACE (angiotensin-converting enzyme) inhibitor/angiotensin II receptor blockers, and mineralocorticoid receptor antagonists (MRAs) contributed 2 points for therapy initiation plus 2 points multiplied by the proportion of target daily dose achieved, while angiotensin receptor–neprilysin inhibitors contributed 4 points for initiation plus 2 points multiplied by the target daily dose. This score reflects the goals of both initiation of therapy and dose uptitration, which are noted in the ACC/AHA Clinical Performance Measures for HF.¹⁰ We have also previously demonstrated the association between this score and subsequent clinical outcomes.⁹ We did not include sodium-glucose cotransporter 2 inhibitors in the primary outcome given the majority of the study period preceded their approval for HF. The total score was a sum of these components, ranging from 0 to 14 (Table S1).

The second primary outcome was the same composite GDMT score estimated under a nonparallel slopes model, which relaxed the assumption of perfectly parallel pre-enrollment trends between GWTG-HF and non-GWTG-HF hospitals. We designated both the parallel and nonparallel slopes specifications as coprimary to provide readers with a comparison of estimates under each assumption.

For each drug class, we identified the last prescription fill within 90 days postdischarge. We used the 90-day window postdischarge because shorter time windows may fail to capture medications in which the patient still has the remainder of an active medication filled before admission. This approach does potentially incorporate postdischarge GDMT optimization in addition to inpatient GDMT optimization; however, prior studies have illustrated very low rates of postdischarge medication optimization.¹¹ We also evaluated the GDMT score at 180 days postdischarge. For patients who died or were rehospitalized during the time window, we still used the last medication fill.

Secondary outcomes included the composite GDMT score for quadruple therapy, including sodium-glucose cotransporter

2 inhibitor, binary indicators for the use of each GDMT medication class, and binary indicators for use of $\geq 50\%$ of target dose for β -blockers and renin-angiotensin-aldosterone system inhibitors (RASIs).¹⁰ Finally, we measured clinical outcomes of HF readmission, all-cause readmission, and death at 30 days, 90 days, and 1-year postdischarge. HF readmission was identified as an acute-care hospitalization with a principal diagnosis of HF.

Study Variables

Patient-level predictor variables included sociodemographic and clinical characteristics. Sociodemographic variables included age, sex, race and ethnicity, rural/urban classification, and CDC social vulnerability index based on zip code. We identified medical comorbidities using a 1-year lookback period before hospitalization based on *International Classification of Diseases, Ninth Revision* and *International Classification of Diseases, Tenth Revision* diagnostic codes (Table S2). Comorbidities included cardiovascular disease (prior HF, myocardial infarction, cerebrovascular disease/transient ischemic attack, and peripheral artery disease) and noncardiac disease (cancer, chronic kidney disease, chronic obstructive pulmonary disease, diabetes, liver disease, and frailty). Hospital-level characteristics included geographic region, bed size, teaching status, and Centers for Medicare and Medicaid Services quality (range, 1–5). For analyses evaluating the GDMT score and individual medication classes, we accounted for the availability of newer therapies over the study period by including indicator variables for periods after angiotensin receptor–neprilysin inhibitor approval (2015) and sodium–glucose cotransporter 2 inhibitor approval for HFrEF (2020).

Statistical Analysis

We compared sociodemographic, hospitalization characteristics, comorbidities, and preadmission HF medications for patients stratified by hospital GWTG-HF participation. In addition, we compared hospital characteristics between hospitals that participated in GWTG-HF and those that never participated during the study period. Given the large sample size, we evaluated the magnitude of differences between groups using standardized mean differences (SMDs). SMD of 0.2, 0.5, and 0.8 were considered small, medium, and large differences, respectively.¹² We also compared groups with χ^2 tests for categorical variables and *t* tests for continuous variables. Missing covariate data were handled using a missing indicator category in adjusted analyses.

We used a mixed-effects model with a random intercept for the discharging hospital to account for baseline outcome levels for each hospital and a continuous measure of each quarter-year in the study period to capture secular trends. To account for within-patient correlation from multiple hospitalizations per patient, we estimated cluster-robust standard errors at the patient level. For the primary analysis, we evaluated a level change (immediate change in the temporal trend) in the outcome after a hospital joined the GWTG-HF registry. We tested whether there were significant differences between temporal trends in GDMT prescription for non-GWTG-HF hospitals and GWTG-HF hospitals before enrolling in GWTG-HF. Because we found no significant difference, we assumed parallel preintervention trends for the primary analysis (Table S3). We then

repeated the primary analysis while modeling different baseline slopes for GWTG-HF and non-GWTG-HF hospitals before registry participation.

To evaluate the potential for bias related to staggered adoption of GWTG-HF participation (eg, early versus late program participation), we performed a Goodman-Bacon decomposition to quantify the contribution of different comparison groups to the overall treatment effect estimate (Table S4). We then conducted a sensitivity analysis using the Callaway-Sant'Anna estimator, which compared participating hospitals only with hospitals that had not yet joined or never joined GWTG-HF.¹³

For the GDMT score, a continuous outcome, we used mixed-effects linear regression. Residuals from the primary model were approximately symmetrical and centered at zero, supporting this approach. For individual medications, readmission, and mortality, we used mixed-effects logistic regression. We adjusted all models for patient age, sex, race and ethnicity, rurality, CDC social vulnerability index quartile, comorbidities, and seasonal variation.

We performed multiple sensitivity analyses. First, we analyzed differences in the GDMT score and clinical outcomes without adjustment for patient characteristics and without the hospital-level random intercept. Second, we repeated the analysis without the hospital-level intercept with adjustment for hospital characteristics. Third, we repeated the primary analysis while excluding those who died before the 90-day follow-up. Lastly, we restricted the primary analysis to index hospitalizations only. We applied a *P* value threshold of <0.05 as statistically significant. The study was approved by the institutional review board.

RESULTS

Among 1274863 Medicare beneficiaries with parts A and D hospitalized for HFrEF, 53.5% ($n=681\,935$) were treated at hospitals that never participated in GWTG-HF and 9.6% ($n=122\,554$) at hospitals that joined GWTG-HF during the study period (Figure 2). The remaining patients were excluded because their hospital had withdrawn from the GWTG-HF registry before the hospitalization (17.6%, $n=224\,025$) or their hospital joined the GWTG-HF registry before the study period (19.3%, $n=246\,349$).

Patient sociodemographic and clinical data are presented in Table 1. The average age was 77 years, and about 42.8% were female. The majority of patients were White (69.0%), followed by Black (19.6%), Hispanic or Latino (8.4%), Asian or Pacific Islander (1.6%), and American Indian or Alaskan Native (0.5%). Nearly 1 in 5 patients resided in rural areas, 21.6% of patients in non-GWTG-HF hospitals compared with 10.2% in GWTG-HF hospitals ($P<0.001$; SMD, 0.3). Patients hospitalized in non-GWTG hospitals were more likely to reside in communities with moderate-high social vulnerability (CDC social vulnerability index, 0.51–0.75; 27.9% versus 23.0%) compared with GWTG-HF hospitals, but the proportion at the highest social vulnerability was similar (CDC social vulnerability index, 0.76–1.00; 31.8% versus

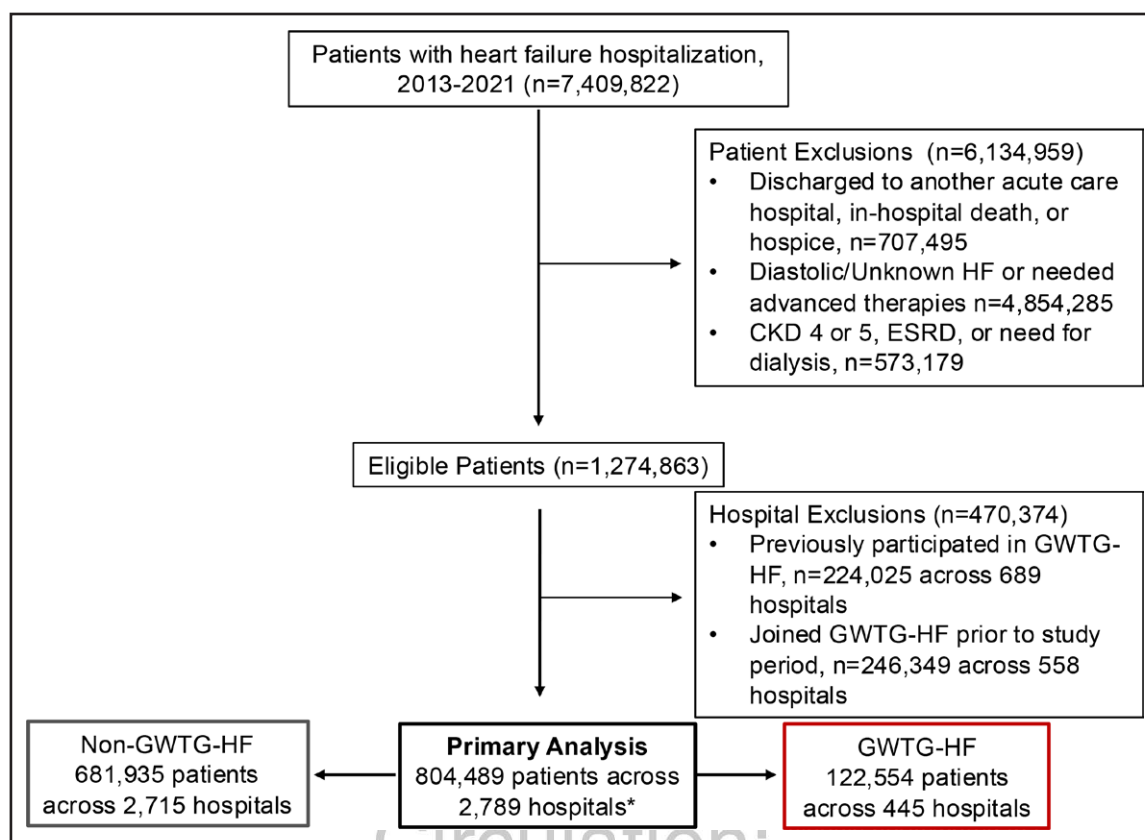


Figure 2. Inclusion and exclusion criteria.

Inclusion and exclusion criteria for primary analysis. *Hospitals could contribute to both groups depending on their Get With The Guidelines–Heart Failure (GWTG-HF) enrollment status during the study period.

30.9%). Cardiac comorbidities were common in both groups, including hypertension (96.5%), coronary artery disease (79.5%), and atrial fibrillation/flutter (62.4%). Noncardiovascular disease was also similar across groups and highly prevalent, such as diabetes (56.6%), chronic kidney disease (56.4%), frailty (21.1%), and liver disease (19.9%).

Baseline HF therapies in the 90 days before hospitalization were similar across both groups with an unadjusted median GDMT score of 3.0 (interquartile range [IQR], 0–5.5; SMD, 0.1). Rates of β -blockers (63.8% versus 62.6%), RASI (51.8% versus 52.4%), and MRA (17.4% versus 16.3%) were also similar between GWTG-HF and non-GWTG-HF hospitals (SMDs of 0.02, 0.01, and 0.03, respectively).

Hospital characteristics are presented in Table 2. There were 445 hospitals that had joined the GWTG-HF registry during the study period and 2344 hospitals that did not join the registry during the study period. Figure S1 displays the overall pattern of hospitals joining the GWTG-HF registry over time. Non-GWTG-HF hospitals were more likely to be small hospitals (<100 beds; 42.3% versus 13.3%), whereas GWTG-HF hospitals were more likely to be large (>400 beds; 25.4% versus 8.2%). GWTG-HF hospitals were more likely to be teaching hospitals (74.4% versus 37.3%; SMD, 0.87).

GDMT Fills After Discharge

Based on Medicare part D prescription fills at 90 days after hospitalization, the median GDMT score increased to 4.0 (IQR, 2.3–6.5) in non-GWTG-HF hospitals and 4.5 (IQR, 2.3–7.0) in GWTG-HF hospitals at 90 days postdischarge ($P<0.001$). At 180 days postdischarge, the median GDMT score increased to 4.8 (IQR, 2.3–7.0) in non-GWTG hospitals and 5.1 (IQR, 2.5–7.8) in GWTG-HF hospitals ($P<0.001$; Table 3). Figure 3 displays the average postdischarge GDMT score across hospitals per month, relative to hospital enrollment in the GWTG-HF registry.

We evaluated the parallel trends assumption by comparing preenrollment temporal trends in GDMT score between hospitals that eventually joined the GWTG-HF registry and those that never joined during the study period. We found no significant differential preenrollment trend between groups (Table S3), supporting the parallel trends assumption underlying our primary analysis. Figure 4 displays the preenrollment trajectories of the GDMT score, highlighting the broadly parallel trend between non-GWTG-HF hospitals and hospitals that joined GWTG-HF in the future.

In the primary adjusted analysis, hospitalization at a GWTG-HF hospital was associated with a significantly

Table 1. Sociodemographic Data, Clinical Characteristics, and Preadmission HF Therapies Stratified by Patients Hospitalized in GWTG-HF Versus Non-GWTG-HF Participating Hospitals

	Non-GWTG-HF	GWTG-HF	P value	SMD
Patients	681 935	122 554		
Age (IQR), y	77 (69–84)	76 (69–84)	<0.0001	0.0233
Female sex, n (%)	293 292 (43.0%)	51 134 (41.7%)	<0.0001	0.0260
Race and ethnicity, n (%)			<0.0001	0.1447
American Indian/Alaska Native	3328 (0.4%)	345 (0.28%)		
Asian/Pacific Islander	9855 (1.5%)	2754 (2.3%)		
Black	129 588 (19.0%)	27 928 (22.8%)		
Hispanic/Latino	56 526 (8.3%)	11 324 (9.2%)		
Non-Hispanic White	476 243 (69.8%)	78 450 (64.0%)		
Unknown	6395 (0.94%)	1753 (1.4%)		
Rural, n (%)	147 174 (21.6%)	12 516 (10.2%)	<0.0001	0.3148
CDC SVI			<0.0001	0.2287
0.00–0.25	118 293 (17.4%)	22 163 (18.1%)		
0.26–0.50	146 377 (21.5%)	34 019 (27.8%)		
0.51–0.75	190 489 (27.9%)	28 196 (23.0%)		
0.76–1.00	217 026 (31.8%)	37 850 (30.9%)		
Length of stay, median (IQR)	4 (2–6)	4 (3–7)	<0.0001	0.0863
Comorbidities				
Alcohol use disorder	45 962 (6.7%)	9001 (7.3%)	<0.0001	0.0236
Atrial fibrillation/flutter	426 244 (62.5%)	75 611 (61.7%)	<0.0001	0.0167
Cancer	109 921 (16.1%)	19 530 (15.9%)	0.1082	0.0050
CVA/TIA	220 196 (32.3%)	36 242 (29.6%)	<0.0001	0.0588
Chronic kidney disease	382 705 (56.1%)	70 923 (57.9%)	<0.0001	0.0354
COPD	371 801 (54.5%)	60 199 (49.1%)	<0.0001	0.1083
Coronary artery disease	543 832 (79.8%)	95 388 (77.8%)	<0.0001	0.0469
Depression	192 408 (28.2%)	32 232 (26.3%)	<0.0001	0.0430
Diabetes	386 378 (56.7%)	68 865 (56.2%)	0.0024	0.0094
Drug use disorder	46 923 (6.9%)	9351 (7.6%)	<0.0001	0.0289
Frailty	146 114 (21.4%)	23 918 (19.5%)	<0.0001	0.0474
Hypertension	657 709 (96.5%)	118 394 (96.6%)	0.0057	0.0086
Liver disease	137 750 (20.2%)	22 582 (18.4%)	<0.0001	0.0449
Prior HF	501 314 (73.5%)	87 792 (71.6%)	<0.0001	0.0421
PAD	305 030 (44.7%)	59 006 (48.2%)	<0.0001	0.0686
Charlson comorbidity index; median (IQR)	7 (5–9)	7 (5–9)	<0.0001	0.0185
Medications 90 d before admission				
β-Blocker	427 106 (62.6%)	78 151 (63.8%)	<0.0001	0.0236
RAASI	357 361 (52.4%)	63 441 (51.8%)	<0.0001	0.0128
ARNI*	29 118 (6.0%)	8553 (7.4%)	<0.0001	0.0552
MRA	110 874 (16.3%)	21 339 (17.4%)	<0.0001	0.0308
SGLT2i†	4179 (2.7%)	1514 (3.1%)	<0.0001	0.0253
GDMT score, median (IQR)	3 (0–5.5)	3 (0–6.0)	<0.0001	0.0609

ARNI indicates angiotensin receptor–neprilysin inhibitor; CDC SVI, Centers for Disease Control and Prevention Social Vulnerability Index; COPD, chronic obstructive pulmonary disease; CVA, cerebrovascular disease; GDMT, guideline-directed medical therapy; GWTG-HF, Get With The Guidelines–heart failure; IQR, interquartile range; MRA, mineralocorticoid receptor antagonist; PAD, peripheral artery disease; RAASI, renin-angiotensin-aldosterone system inhibitor; SGLT2i, sodium-glucose cotransporter 2 inhibitor; SMD, standardized mean difference; and TIA, transient ischemic attack.

*Patients hospitalized after August, 2015.

†Patients hospitalized after October, 2019.

Table 2. Hospital Characteristics Stratified by GWTG-HF Participation During Study Period

	Non-GWTG-HF	GWTG-HF	P value	SMD
No. of hospitals, n	2344	445		
No. of beds, n (%)			<0.0001	0.9286
<100	992 (42.3%)	59 (13.3%)		
100–400	967 (41.3%)	272 (61.1%)		
>400	191 (8.2%)	113 (25.4%)		
Missing	194 (8.3%)	1 (0.22%)		
Region, n (%)			<0.0001	0.5511
Northeast	300 (12.8%)	91 (20.5%)		
Midwest	508 (21.7%)	112 (25.2%)		
South	972 (41.5%)	127 (28.5%)		
West	370 (15.8%)	114 (25.6%)		
Missing	194 (8.3%)	1 (0.22%)		
Teaching hospital, n (%)			<0.0001	0.7660
Yes	959 (37.3%)	331 (74.4%)		
No	1191 (50.8%)	113 (25.4%)		
Missing	194 (8.3%)	1 (0.22%)		
CMS quality rating			<0.0001	0.7738
1	136 (5.8%)	31 (7.0%)		
2	369 (15.7%)	95 (21.4%)		
3	525 (22.4%)	133 (29.9%)		
4	447 (19.1%)	99 (22.3%)		
5	223 (9.5%)	77 (17.3%)		
Missing	644 (27.5%)	10 (2.3%)		

Hospitals that participated exclusively in GWTG-HF or as a non-GWTG-HF hospital during the study period were included. GWTG-HF indicates Get With The Guidelines–Heart Failure; and SMD, standardized mean difference.



higher 90-day GDMT score compared with hospitalization at a non-GWTG-HF hospital (+0.15 points [95% CI, 0.12–0.20]; $P<0.001$) at 90 days. The difference was similar at 180 days (+0.16 points [95% CI, 0.12–0.20]; $P<0.001$). In the coprimary analysis using a nonparallel slopes model, the level difference was attenuated to +0.08 ([95% CI, 0.03–0.15]; $P<0.001$) and +0.07 ([95% CI, 0.01–0.13]; $P=0.005$) at 90 and 180 days, respectively (Table S3).

The Callaway-Sant'Anna sensitivity analysis yielded an absolute treatment effect of +0.16 ([95% CI, −0.18 to 0.49]; Table S4). In a sensitivity analysis, in which we adjusted for hospital characteristics rather than the baseline hospital outcome, we found similar differences in GDMT score between GWTG-HF and non-GWTG-HF hospitals (+0.13 points [95% CI, 0.11–0.15]; $P<0.001$). When excluding patients who died before the 90-day follow-up, the difference in GDMT score between GWTG-HF and non-GWTG-HF hospitals was +0.17 points ([95% CI, 0.12–0.22]; $P<0.001$) at 90 days. Lastly, we obtained similar results when restricting the primary analysis to index hospitalizations only (Table S5).

After including sodium-glucose cotransporter 2 inhibitor in the GDMT score, the median postdischarge score at 90 days was 4.0 (IQR, 2.3–6.5) in non-GWTG hospitals

and 4.5 (IQR, 2.3–7.0) in GWTG hospitals ($P<0.001$). After adjustment, hospitalization at a GWTG-HF hospital was associated with a significantly higher GDMT score at 90 days postdischarge compared with hospitalization at a non-GWTG-HF hospital (+0.18 points [95% CI, 0.13–0.23]; $P<0.001$).

Patients treated at GWTG-HF hospitals were more likely to have filled β -blockers (OR, 1.07 [95% CI, 1.06–1.09]), RASI (OR, 1.02 [95% CI, 1.01–1.03]), and MRA (OR, 1.13 [95% CI, 1.11–1.15]) at 90 days after hospitalization compared with those in non-GWTG-HF hospitals. There was no significant difference in angiotensin receptor–neprilysin inhibitor fills (OR, 1.02 [95% CI, 0.99–1.04]). Patients in GWTG-HF hospitals were more likely to achieve $\geq 50\%$ of target β -blocker doses (OR, 1.08 [95% CI, 1.06–1.10]) but were less likely to achieve $\geq 50\%$ of target RASI therapy (OR, 0.95 [95% CI, 0.93–0.97]) at 90 days (Figure 5).

In analyses accounting for nonparallel trends before GWTG-HF participation, these associations were attenuated. The likelihood of MRA fills remained higher in GWTG-HF hospitals (OR, 1.07 [95% CI, 1.04–1.10]), whereas the likelihood of RASI decreased slightly (OR, 0.98 [95% CI, 0.95–1.00]). Other differences were no longer statistically significant (Table S6).

Table 3. Unadjusted Rates of Composite GDMT Score, Individual HF Medications, and Clinical Outcomes Stratified by GWTG-HF Participation

	Non-GWTG-HF	GWTG-HF	P value
Composite GDMT score, median (IQR)			
GDMT score,* 90 d	4.0 (2.3–6.5)	4.5 (2.3–7.0)	<0.001
GDMT score,* 180 d	4.8 (2.3–7.0)	5.1 (2.5–7.8)	<0.001
GDMT including SGLT2i, 90 d	4.0 (2.3–6.5)	4.5 (2.3–7.0)	<0.001
GDMT including SGLT2i, 180 d	4.8 (2.3–7.0)	5.2 (2.5–8.0)	<0.001
Posthospitalization HF therapies 90 d, n (%)			
β-Blocker, any dose	484 014 (71.0%)	90 130 (73.5%)	<0.001
β-Blocker (≥50% target dose)	160 991 (23.6%)	32 021 (26.1%)	<0.001
RASI, any dose	385 308 (56.5%)	69 221 (56.5%)	0.896
RASI, ≥50% target dose	107 553 (15.8%)	17 478 (14.3%)	<0.001
ARNI, any dose	49 676 (7.3%)	14 414 (11.8%)	<0.001
MRA, any dose	181 959 (26.7%)	36 395 (29.7%)	<0.001
Posthospitalization HF therapies, 180 d, n (%)			
β-Blocker, any dose	525 448 (77.1%)	97 356 (79.4%)	<0.001
β-Blocker, ≥50% target dose	186 712 (27.4%)	37 310 (30.4%)	<0.001
RASI, any dose	424 583 (62.3%)	76 178 (62.2%)	0.494
RASI, ≥50% target dose	125 805 (18.5%)	20 809 (17.0%)	<0.001
ARNI, any dose	60 392 (8.9%)	17 341 (14.2%)	<0.001
MRA, any dose	209 838 (30.8%)	41 832 (34.1%)	<0.001
Clinical outcomes, n (%)			
All-cause hospitalization, 30 d	164 607 (24.1%)	29 834 (24.3%)	0.122
All-cause hospitalization, 90 d	293 875 (43.1%)	52 835 (43.1%)	0.910
All-cause hospitalization, 1 y	451 376 (66.2%)	81 047 (66.1%)	0.689
HF hospitalization, 30 d	75 574 (11.1%)	13 814 (11.3%)	0.052
HF hospitalization, 90 d	146 806 (21.5%)	26 931 (22.0%)	<0.001
HF hospitalization, 1 y	242 463 (35.6%)	44 358 (36.2%)	<0.001
Death, 30 d	46 087 (6.8%)	7553 (6.2%)	<0.001
Death, 90 d	114 693 (16.8%)	19 500 (15.9%)	<0.001
Death, 1 y	259 119 (38.0%)	44 690 (36.5%)	<0.001

ARNI indicates angiotensin receptor–neprilysin inhibitor; GDMT, guideline-directed medical therapy; GWTG-HF, Get With The Guidelines–Heart Failure; HF, heart failure; MRA, mineralocorticoid receptor antagonist; RASI, renin-angiotensin system inhibitor; RAASi, renin-angiotensin aldosterone system inhibitor; and SGLT2i, sodium-glucose cotransporter 2 inhibitor.

*Reflects use and dose intensity of β-blockers, RASI, and MRA.

Clinical Outcomes

At 30 days postdischarge, HF readmission was 11.3% for patients hospitalized at GWTG-HF hospitals and 11.1% for patients hospitalized at non-GWTG-HF hospitals ($P=0.05$). The proportion of all-cause and HF readmissions at 90 days and 1 year was also similar (Table 3). After adjustment, GWTG-HF participation was not associated with differences in HF readmissions or all-cause readmissions (Figure 5; Table S7).

At 30 days postdischarge, all-cause mortality was 6.2% for patients hospitalized at GWTG-HF hospitals and 6.8% for patients hospitalized at non-GWTG-HF hospitals ($P<0.001$). At 90 days, mortality was 15.9% versus 16.8% ($P<0.001$) and at 1 year, 36.5% versus 38.0% ($P<0.001$). After adjustment, GWTG-HF participation

was associated with significantly lower postdischarge mortality at 30 days (OR, 0.95 [95% CI, 0.92–0.98]) and 1 year (0.97 [95% CI, 0.95–1.00]), both $P<0.05$, but missed statistical significance at 90 days (OR, 0.98 [95% CI, 0.95–1.00]; $P=0.07$). In the analysis assuming nonparallel trends before GWTG-HF participation, this difference was attenuated (Table S7) and no longer statistically significant.

The Cox proportional hazards models evaluating time to death and first HF readmission through 1 year were consistent with the logistic regression analyses. GWTG-HF participation was associated with a lower risk of death in the parallel slopes model (HR, 0.97 [95% CI, 0.95–0.99]), but this association was attenuated and no longer significant in the nonparallel slopes model (HR, 0.99 [95% CI, 0.96–1.02]). GWTG-HF participation was

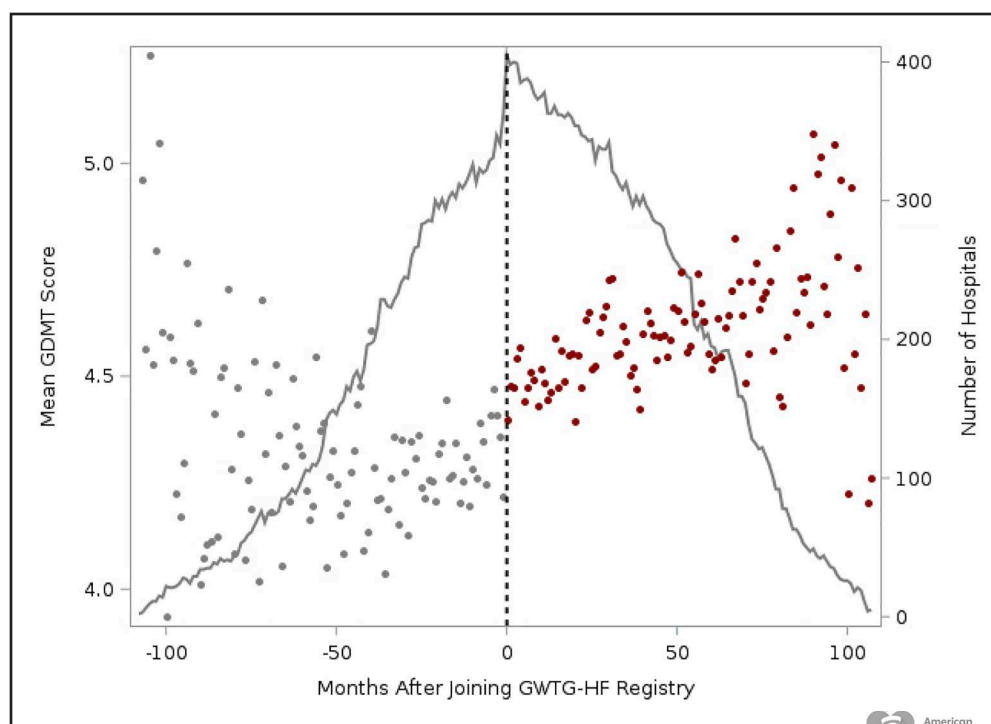


Figure 3. Temporal trends in posthospitalization mean guideline-directed medical therapy (GDMT) scores relative to Get With The Guidelines–Heart Failure (GWTG-HF) enrollment.

Scatter plot representing the mean posthospitalization GDMT score across hospitals per month. The vertical dashed line indicates the time of GWTG-HF enrollment (month 0), with observations before enrollment (gray) and after enrollment (red). The gray line represents the number of hospitals that contribute data at each time point (right y axis), reflecting the staggered enrollment of hospitals over the study period.

not associated with differences in HF readmission in either model (Table S8).

DISCUSSION

Among Medicare beneficiaries hospitalized with HF, we found that hospital participation in the GWTG-HF program was associated with significantly greater use of GDMT, including higher odds of postdischarge treatment with β -blockers, RASI, and MRA. These findings expand on prior studies demonstrating improved process of care metrics among GWTG-HF hospitals by evaluating within-hospital changes after registry enrollment and provide a contemporary evaluation of the program's role in promoting GDMT for HF.^{4,14}

Patient demographics, comorbidities, and baseline therapy were overall similar between patients admitted to hospitals that joined the GWTG-HF registry and those that did not. This is an important observation that supports the value of the GWTG-HF registry in analyzing temporal trends in acute HF care in the United States. Nevertheless, there are certain differences in patient and hospital characteristics. Non-GWTG-HF hospitals served twice as many rural patients, a population with higher HF mortality risk and greater barriers to accessing care that could potentially limit HF care and outcomes.¹⁵ GWTG-HF hospitals also tended to be larger (>400 beds), suggesting possible structural

advantages that may have facilitated participation in the program. Targeting recruitment of GWTG-HF to smaller, rural hospitals is an important opportunity to improve quality in populations that already face significant barriers to optimal HF treatment. The AHA has aimed to address this difference via the Rural Health Care Outcomes Accelerator, which aims to increase participation of rural hospitals at no cost in the GWTG registries.¹⁶ Additionally, GWTG-HF hospitals more frequently had an Accreditation Council for Graduate Medical Education training program and higher Centers for Medicare and Medicaid Services quality scores, although Centers for Medicare and Medicaid Services quality ratings were more often missing among non-GWTG-HF hospitals (27.5%), limiting interpretation. Our analysis accounted for differences in hospital characteristics by adjusting for baseline hospital performance, and we performed a sensitivity analysis adjusting for these characteristics that demonstrated similar findings.

Although we found a significant improvement in the GDMT composite score after GWTG-HF participation in the parallel slopes model, the estimated effect was attenuated in the nonparallel slopes model. Given that our findings were sensitive to modeling assumptions, these findings may suggest that hospitals opting into GWTG-HF were already on a trajectory of quality improvement before enrollment. Participation in GWTG-HF may still provide the structure, benchmarking, and

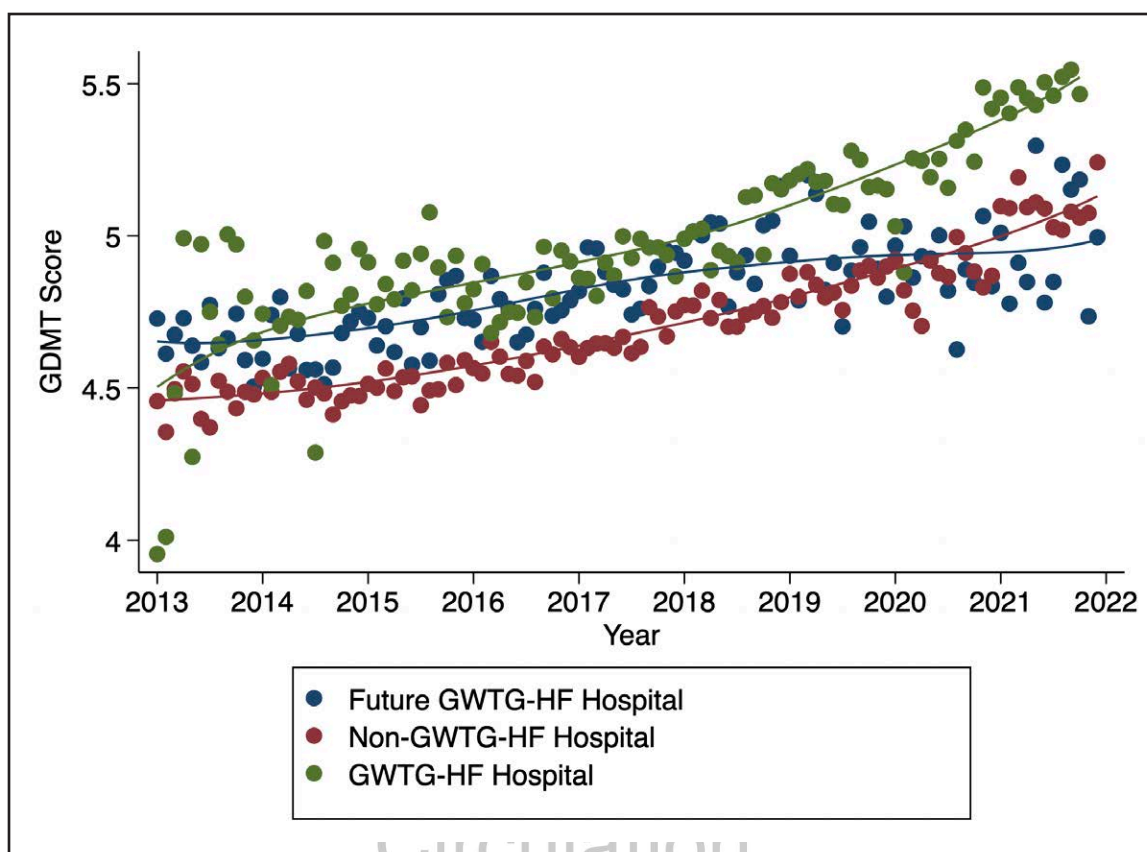


Figure 4. Postdischarge guideline-directed medical therapy (GDMT) scores stratified by Get With The Guidelines–Heart Failure (GWTG-HF) hospital participation status.

Each point represents the mean 90 days postdischarge GDMT score for an individual hospital stratified by participation status: future GWTG-HF hospitals (blue), non-GWTG-HF hospitals (red), and actively enrolled GWTG-HF hospitals (green). Superimposed lines represent fitted loess lines for each group.

reporting mechanisms to support and sustain quality improvement efforts in HF care.

We found that the magnitude of GDMT improvement associated with GWTG-HF participation was modest. Several factors likely contributed to this finding. First, patient-level barriers—including hypotension, frailty, chronic kidney disease, medication intolerance, and out-of-pocket costs—may limit GDMT implementation but were incompletely captured in administrative claims data. Inclusion of patients with contraindications or socioeconomic barriers may have therefore diluted treatment effects. Second, during much of the study period, GWTG-HF performance measures emphasized therapies such as β -blockers and ACE inhibitors/angiotensin II receptor blocker therapy that had already achieved relatively high adoption rates. By contrast, MRAs remained underused because of concerns about renal function and hyperkalemia, although SGLT2 inhibitors were incorporated into HF guidelines only late in the study period. In addition, GWTG-HF primarily measured whether therapies were prescribed rather than titrated to target doses. Together, these factors may have limited the impact on overall GDMT intensity and clinical outcomes.

The modest improvement in GDMT associated with GWTG-HF participation was accompanied by a small reduction in mortality in the parallel slopes model; however, this association was attenuated and no longer significant in the nonparallel slopes model and was not accompanied by a reduction in HF readmission. This may reflect the relatively small magnitude of treatment differences observed, residual confounding, or the fact that hospital-based quality improvement programs influence only one component of longitudinal HF management. Prior work by Steverson et al⁹ demonstrated that higher continuous 4-pillar scores, which informed our GDMT score, were associated with lower mortality risk. The observed improvement in GDMT score in our analysis would likely correspond to a small relative reduction in mortality. Nevertheless, as our analysis evaluated the impact of a hospital-level quality improvement program across >800 000 hospitalizations, even the modest improvements in GDMT intensity may translate into meaningful population-level benefits.

Future efforts should focus on identifying the most effective practices within GWTG-HF, expanding support for under-resourced hospitals, and strengthening

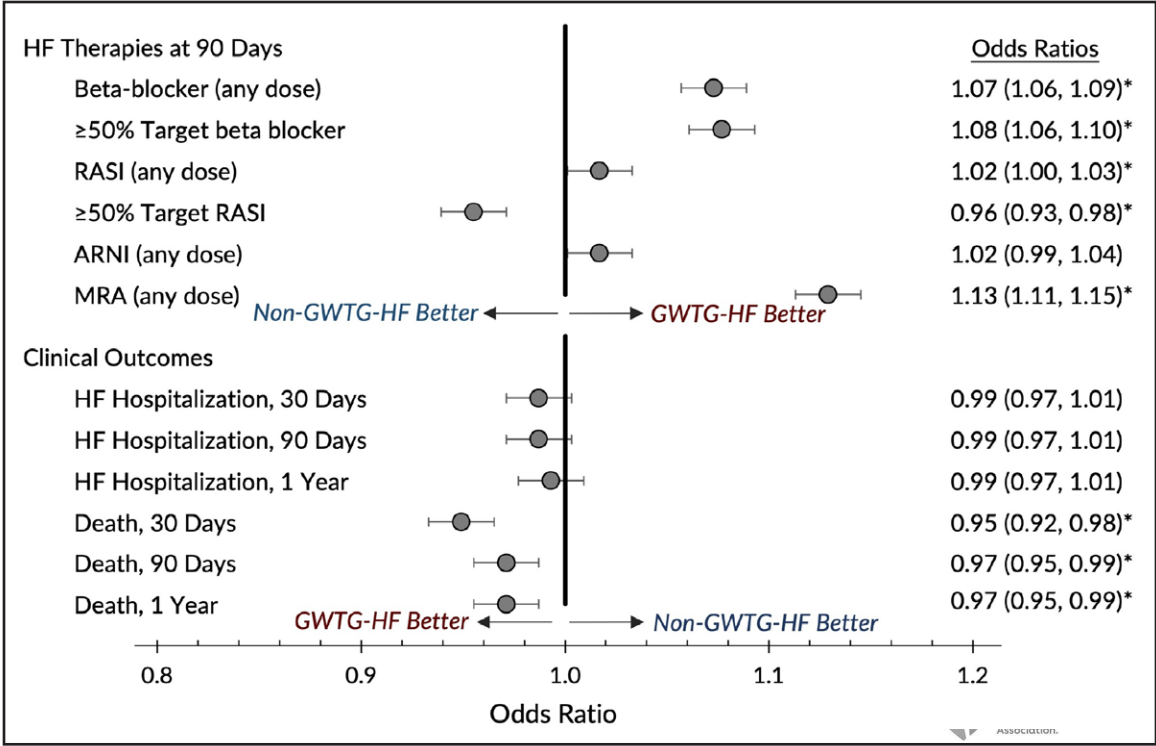


Figure 5. Heart failure (HF) therapies and clinical outcomes. Forest plot illustrating adjusted odds ratios of HF therapy use at 90 days posthospitalization and clinical outcomes among patients treated at Get With The Guidelines–Heart Failure (GWTG-HF) vs non-GWTG-HF hospitals (* $P < 0.05$).

transitions to outpatient care where ongoing medication optimization occurs.¹⁷ The AHA IMPLEMENT-HF initiative illustrates this next phase of quality improvement into the outpatient setting where sustained therapy optimization can have the greatest long-term impact. The program has achieved significant improvements in 4-pillar GDMT and >10% increases in prescriptions for each class of therapy.¹⁸ In addition, recent changes to GWTG-HF Performance Achievement Award criteria now require documentation of prescribing or contraindications for all foundational GDMT classes in eligible patients and may enhance the program's impact in future years.

There are several limitations to this observational study. GWTG-HF hospital participation is voluntary and may be influenced by institutional characteristics, such as greater resource availability, leadership engagement, and a culture of quality improvement that both increase the likelihood of participation and the hospital's capacity to improve care. Our approach accounted for hospital-level differences by adjusting for each hospital's baseline; however, it does not account for changes in hospital characteristics over time that may have influenced both enrollment in GWTG-HF and improvement in outcomes. As such, we cannot determine whether the observed improvements were a direct result of the GWTG-HF program, or if GWTG-HF participation was a marker of hospitals that delivered higher quality care. Nevertheless, our primary analysis focused on change in the composite GDMT score compared with preadmission,

given this is a process metric that is highly under the influence of a treating hospital rather than an outcome measure that is more subject to changes in patient characteristics over time. Finally, administrative claims data do not capture important clinical factors influencing GDMT eligibility, including medication intolerance, contraindications, patient preferences, cost, and device-based HF therapies. Misclassification of HFrEF based on diagnosis coding is also possible. For these reasons, our GDMT rates may differ overall from those observed in analyses of clinical registries, such as the GWTG-HF registry.

CONCLUSIONS

In this national analysis, hospital participation in GWTG-HF was associated with modest improvements in post-discharged GDMT intensity among patients hospitalized with HFrEF, although the magnitude of the association was sensitive to assumptions regarding preenrollment trends. These findings support the role of structured quality programs in driving the uptake of evidence-based HFrEF therapies and highlight opportunities to further strengthen implementation strategies to enhance their clinical impact.

ARTICLE INFORMATION

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Affiliations

Division of Cardiovascular Medicine, Department of Medicine, Cardiovascular Institute, Stanford University School of Medicine, CA (A.V., P.H., A.S.V., J.F.). Division of Cardiology, Department of Medicine, David Geffen School of Medicine at the University of California, Los Angeles (G.C.F., N.B., A.T.S.). Vision of Cardiology, Veterans Affairs Palo Alto Health Care System, CA (P.H., J.F.). Division of Cardiology, Department of Medicine, University of Colorado, Aurora (L.A.A.). Department of Cardiology, Kaiser Permanente San Francisco Medical Center, CA (A.P.A.). Division of Research, Pleasanton, CA (A.P.A.). Department of Cardiology, Keio University School of Medicine, Shinjuku, Japan (S.K.). Division of Cardiology, VA Greater Los Angeles Healthcare System, CA (A.T.S.).

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Supplemental Material

Figure S1
Tables S1–S8
RECORD Checklist

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